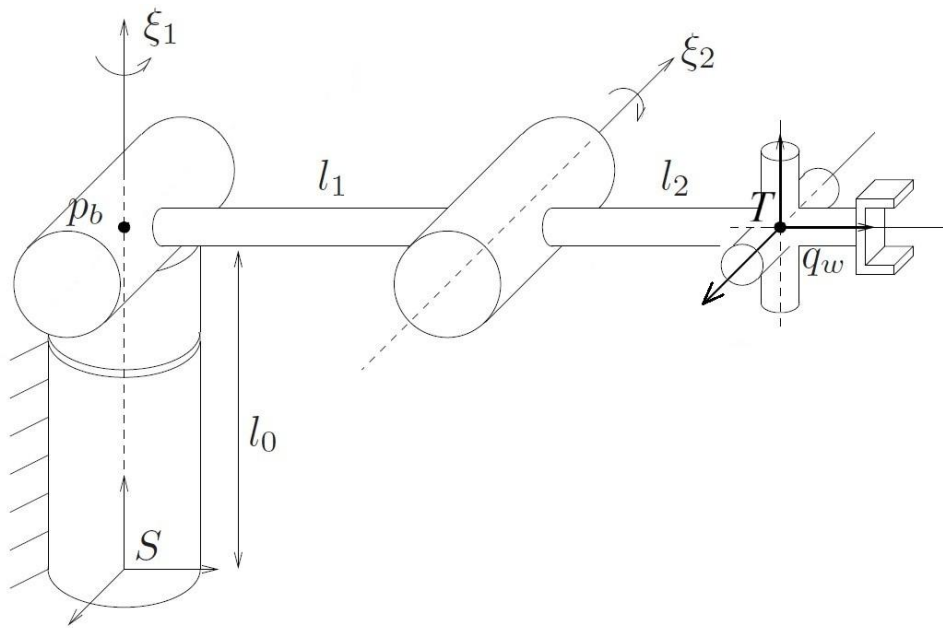


Question 4



```
clearvars, clc
```

```
l0 = 0.5;
l1 = 0.8;
l2 = 0.5;
```

```
r1 = l1/3;
r2 = l2/3;
```

```
g_st0 = [eye(3),[0;l1+l2;l0];zeros(1,3),1]
```

```
g_st0 = 4x4
    1.0000         0         0         0
         0     1.0000         0     1.3000
         0         0     1.0000     0.5000
         0         0         0     1.0000
```

```
pb = [0;0;l0];
q2 = [0;l1;l0];
```

```
omega1 = [0;0;1];
omega2 = [-1;0;0];
```

```
xi1 = [cross(pb,omega1);omega1]
```

```
xi1 = 6x1
     0
     0
     0
     0
     0
     1
```

```
xi2 = [cross(q2,omega2);omega2]
```

```

xi2 = 6×1
    0
   -0.5000
    0.8000
   -1.0000
    0
    0

```

```

xi = [xi1,xi2];

m1 = 70;
m2 = 40;

m2gr2 = m2*9.81*r2;

Ix1 = 4.1;
Iy1 = 0.94;
Iz1 = 5.1;

Ix2 = 2.5;
Iy2 = 0.65;
Iz2 = 2.35;

% as in Eq (4.9) p. 162
calM1 = [m1*eye(3) zeros(3);zeros(3) diag([Ix1 Iy1 Iz1])];
calM2 = [m2*eye(3) zeros(3);zeros(3) diag([Ix2 Iy2 Iz2])];

gsl10 = [eye(3) [0;r1;l0];zeros(1,3),1];
Adgsl10 = adjmg(gsl10);
Adgsl10m1 = inv(Adgsl10);

gsl20 = [eye(3) [0;l1+r2;l0];zeros(1,3),1];
Adgsl20 = adjmg(gsl20);
Adgsl20m1 = inv(Adgsl20);

% Eq (4.28) p. 176
calMp = Adgsl10m1'*calM1*Adgsl10m1;
calMp(1:6,1:6,2) = Adgsl20m1'*calM2*Adgsl20m1;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

beta1 = 1.1; % Nms/rad
beta2 = 0.63; % Nms/rad
beta = [beta1,beta2];

```

The calculation of the matrix of starting values for the index l to be used in applying MLS eq.-s (2.29) & (4.30) is calculated next. For efficient code, this calculation should preferably be done outside a solution loop.

```

lM = zeros(2);
for ii = 1:2
    for jj = 1:2
        lM(ii,jj) = max([ii,jj]);
    end
end

```

```
end
end
```

Consider now a constant applied torque τ with known numerical values.

We wish to solve the manipulator differential equations of motion, MLS eq. (4.47) p. 190:

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + N(\theta, \dot{\theta}) = \tau .$$

This can be re-written as

$$\frac{d}{dt} \begin{bmatrix} \dot{\theta} \\ \theta \end{bmatrix} = \begin{bmatrix} -M(\theta)^{-1}C(\theta, \dot{\theta}) & 0 \\ I & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \theta \end{bmatrix} + \begin{bmatrix} M(\theta)^{-1}(\tau - N(\theta, \dot{\theta})) \\ 0 \end{bmatrix}$$

and then solved with a Runge-Kutta type solution. The Runge-Kutta solution requires a function to calculate the time derivative of the state vector.

```
function dydt = odefcn(~,y,xi,calMp,lM,beta,m2gr2,tau)
    dtheta = y(1:2);
    theta = y(3:4);
    dydt = zeros(4,1);
    [M,C,N] = MCN(xi,calMp,lM,beta,m2gr2,theta,dtheta);

    Mm1 = M\eye(2);
    dydt(1:2,1) = Mm1*(tau - C*y(1:2,1) - N);
    dydt(3:4,1) = y(1:2,1);
end
```

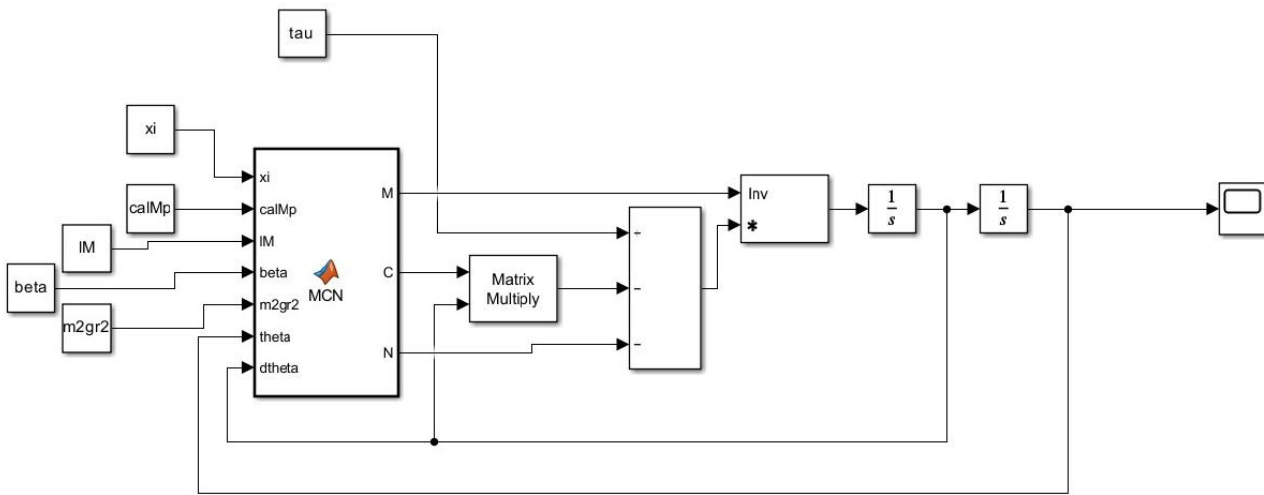
Solution

```
tau = [0; 0]
```

```
tau = 2x1
      0
      0
```

```
% tau = [0.04; 0]
% tau = [0.04; -m2gr2]
% tau = [0.04; -m2gr2 - 0.00001]
% tau = [0.0; -m2gr2]
y0 = zeros(4,1);
tspan = linspace(0,50,400);
%tspan = [0 5];

[t,y] = ode45(@(t,y) odefcn(t,y,xi,calMp,lM,beta,m2gr2,tau), tspan, y0);
```



```
simout = sim(ques4sim)
```

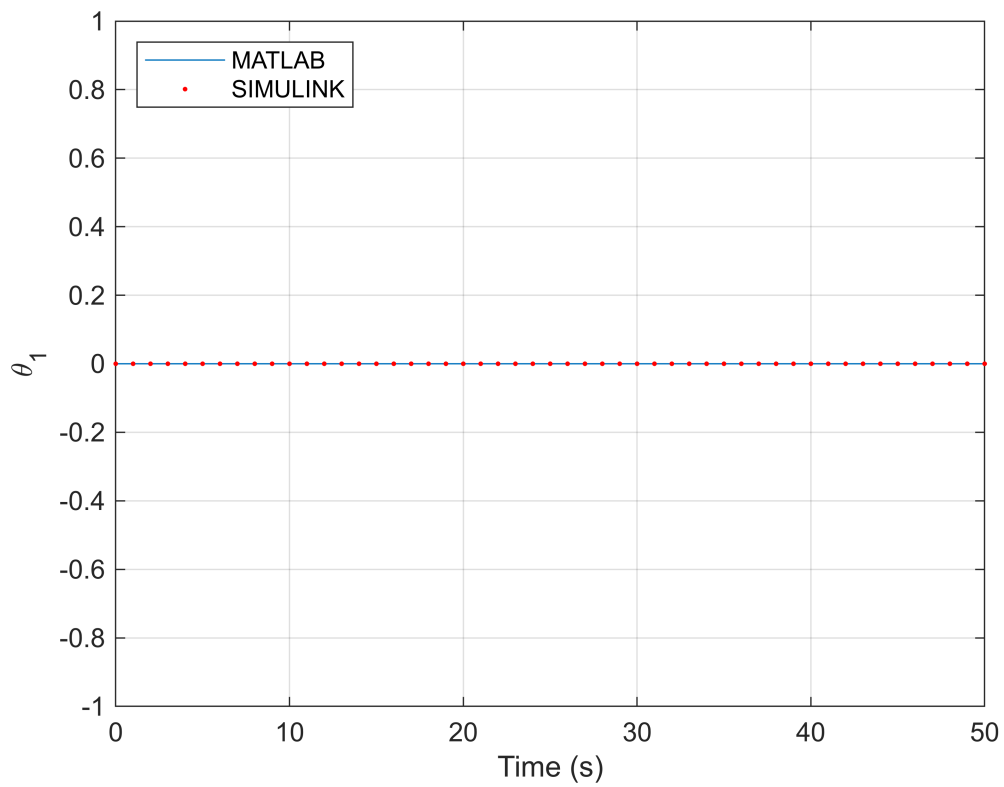
```
simout =
  Simulink.SimulationOutput:

    ScopeData: [1x1 struct]
              tout: [5001x1 double]

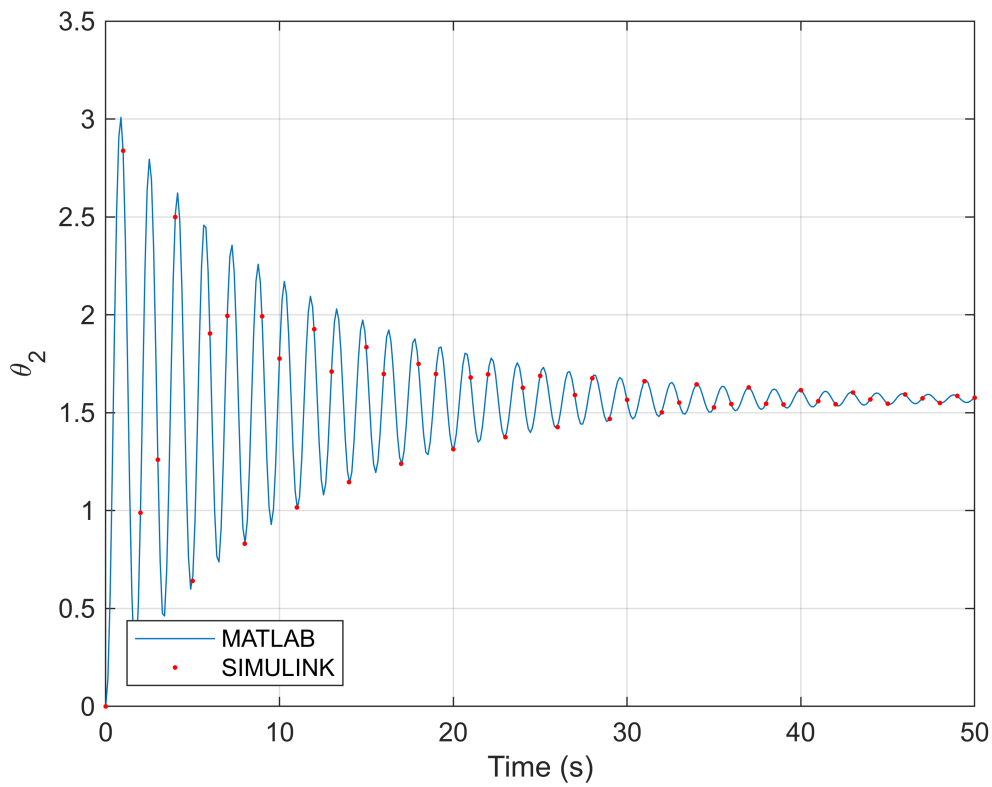
    SimulationMetadata: [1x1 Simulink.SimulationMetadata]
    ErrorMessage: [0x0 char]
```

```
t2 = simout.ScopeData.time;
y2 = simout.ScopeData.signals.values;
y2 = squeeze(y2);
y2 = y2';

figure(1)
plot(t,y(:,3),'DisplayName','MATLAB')
hold on
plot(t2(1:100:end),y2(1:100:end,1),'r.','DisplayName','SIMULINK')
hold off
xlabel('Time (s)')
ylabel('\theta_1')
grid on
legend('location','northwest')
```



```
figure(2)
plot(t,y(:,4),'DisplayName','MATLAB')
hold on
plot(t2(1:100:end),y2(1:100:end,2),'r.','DisplayName','SIMULINK')
hold off
% hold on
% plot([t(1) t(end)],pi/2*[1.0 1.0])
% hold off
xlabel('Time (s)')
ylabel('\theta_2')
grid on
legend('location','southwest')
```



```
Iox2 = Ix2+m2*r2*r2
```

```
Iox2 =  
3.6111
```

```
omega = sqrt(m2*gr2/Iox2)
```

```
omega =  
4.2557
```

```
Ioz1 = (Iz1 + m1*r1*r1 + Iz2 + m2*(l1+r2)^2)
```

```
Ioz1 =  
49.8056
```

```
2*Ioz1/50/50
```

```
ans =  
0.0398
```

```
alpha = tau(1)/Ioz1
```

```
alpha =  
0
```

```
alpha*t(end)^2/2
```

```
ans =  
0
```

Discussion:

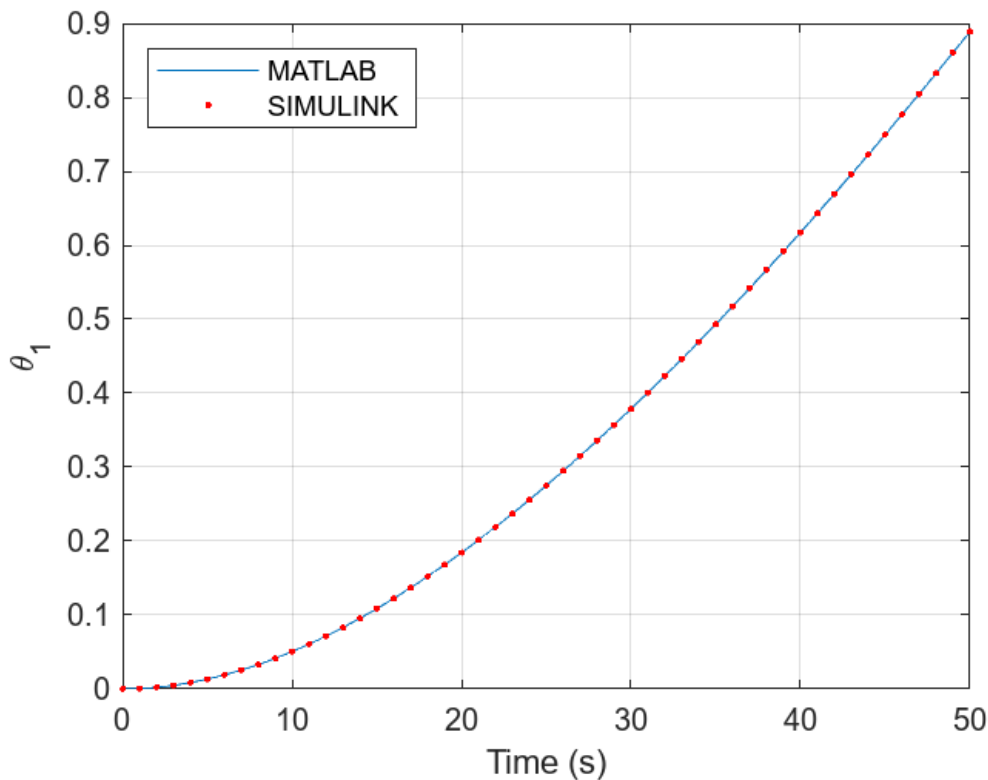
Part (a)

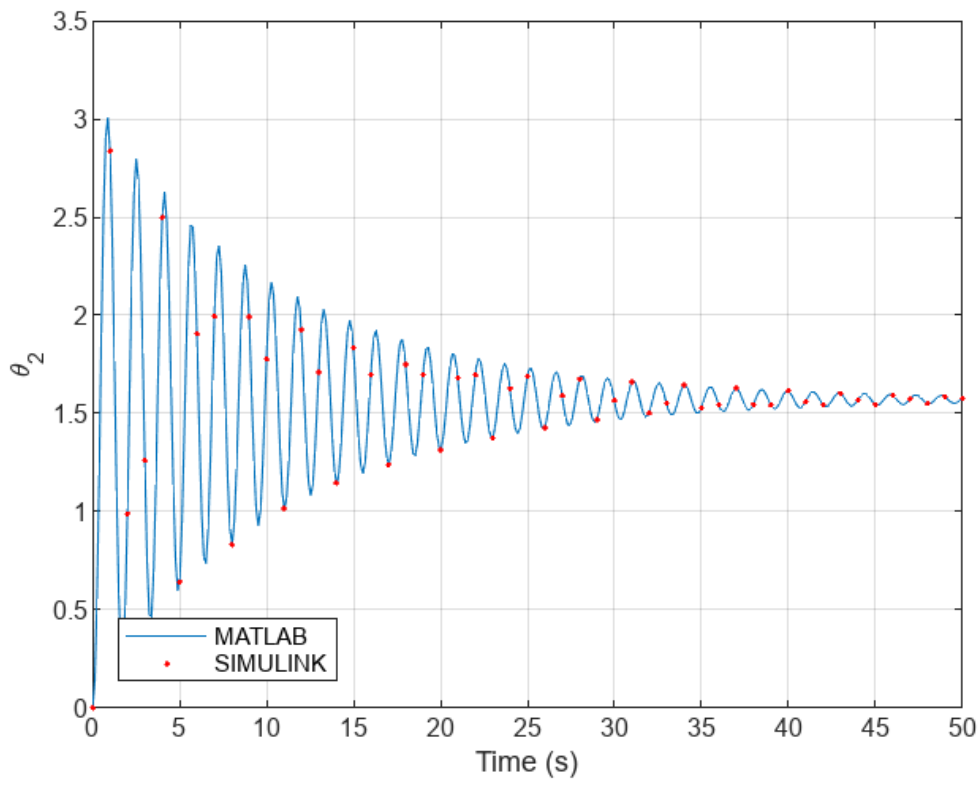
(Figures generated by the code.)

$$\tau = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Because $\tau_1 = 0$, θ_1 is not excited and remains at zero. Even though $\tau_2 = 0$ also, due to its weight, link 2 acts as a compound (not simple) pendulum, starting off from an initial disturbed angle of $\pi/2$ radians relative to its weight vector. (A compound pendulum is one where the mass is not concentrated at the tip of the bar.) It will therefore oscillate, initially with large angles, which means its motion is not described by a linear differential equation. Due to the damping in the bearing, though, the oscillation decays, and after a while (when the maximum angle excursion is less than about $1/6$ radians) one can determine the damped natural frequency from the decaying oscillation period, from the time graph. If one does, one finds excellent agreement with the theoretically predicted undamped natural frequency of a compound pendulum with the parameters of this case.

Part (b)

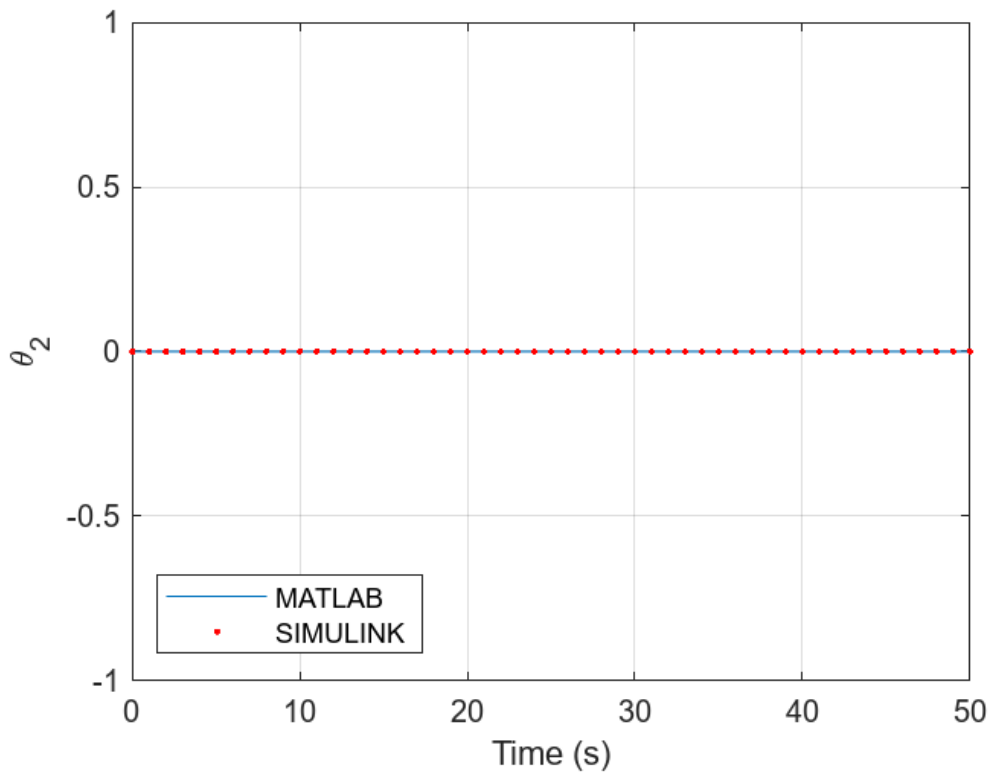
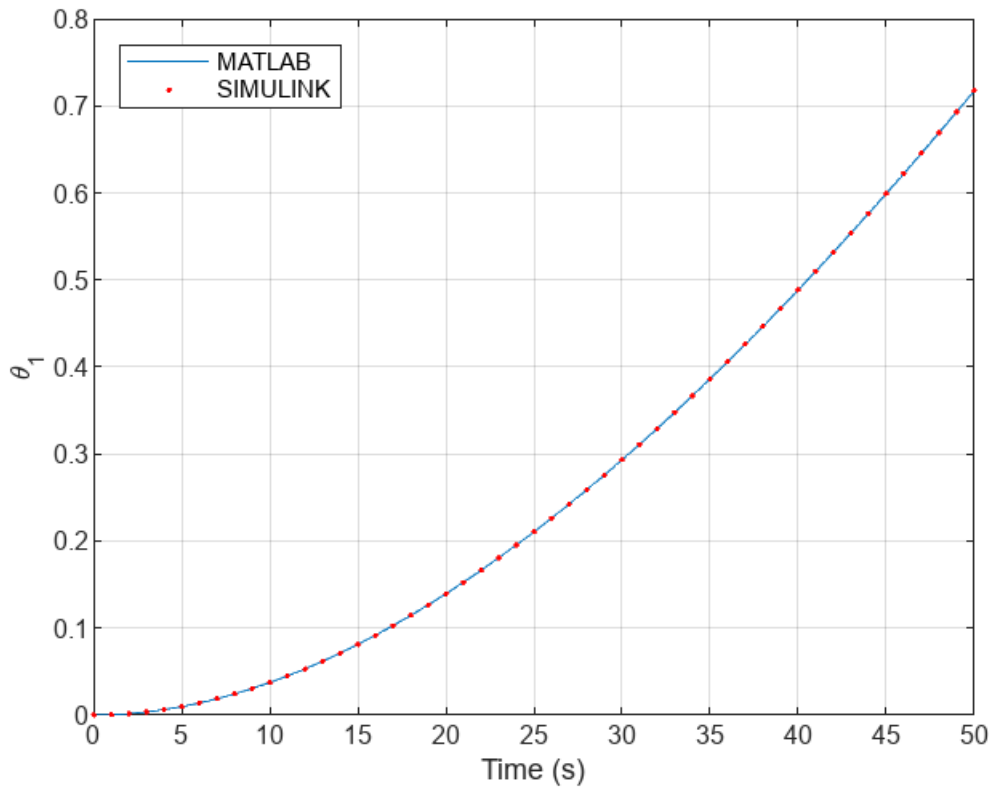




$$\tau = \begin{bmatrix} 0.04 \\ 0 \end{bmatrix}$$

Because $\tau_1 = 0.04$ Nm, θ_1 accelerates and moves in almost a parabolic trajectory, as is expected. As a very rough check, based on a rough estimate of the moment of inertia of the whole manipulator about the ξ_1 axis, one would expect a total θ_1 rotation of about 1 radian after 50 seconds. The graph shows a total rotation of just below 0.9 radians.

Part (c)



$$\tau = \begin{bmatrix} 0 \\ -m_2 g r_2 \end{bmatrix}$$

Because $\tau_1 = 0.04 \text{ Nm}$, θ_1 accelerates and moves in almost a parabolic trajectory, as is expected. Because $\tau_2 = -m_2 g r_2$, which exactly counter the weight of link 2, θ_2 remains zero, as expected. Interestingly, because the centre of mass of link 2 now remains a constant distance from the ξ_1 axis, the effective moment of inertia of link 2 remains at its maximum value, and thus effectively larger than the case in part (b). This seems to be reflected by the slightly lower final value of θ_1 after 50 seconds, of just above 0.7 radians.